

Jiayi Wang (James)

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

August 2025 - present

M.S. in Music Technology; GPA 4/4; Music Informatics Lab; Robotic Musicianship Lab

Occidental College, Los Angeles, CA.

August 2020 - May 2025

B.A. in Music, Emphasis in Music Production; B.A. in Computer Science; Minor in Mathematics

GPA 3.88/4; **Magna Cum Laude**; Dean's list: All semesters to date;

Kurtich Music Grant: four semester of scholarship from 2023 to 2025

PUBLICATIONS & ACADEMIC SERVICE

- A. T. Ma, A. Cammarota, **J. Wang**, A. Smith, C. Z. A. Huang, J. Albert, and A. Lerch, "H2H Music Improv: A Communication Model and Audio-Visual Dataset for Music Improvisation," Proceedings of the International Society for Music Information Retrieval Conference (ISMIR), 2026.
- S. Hudson and **J. Wang**, "Harmonic Divorce or Not? Subjectively Hearing Pedals as Multi-Layered Chords," Music Theory Online, vol. 34, no. 4.
- **J. Wang**, "Cinematic Pedal in Rock and Pop Music," presented at the Occidental College Summer Research Conference, Los Angeles, CA, Aug. 2023.
- **J. Wang**, "Re-evaluating Use Cases in VIVO," presented at the Andrew W. Mellon Summer Faculty Institute Conference, Los Angeles, CA, Aug. 2022.
- **Reviewer**, International Society for Music Information Retrieval Conference (ISMIR), 2026.

INDUSTRY EXPERIENCE

Audio Research Intern, Tencent Music Entertainment (TME), Guangzhou, China

June - August 2026

- Developed MIDI-to-guitar fingering **LLM** model; fine-tuned **T5** and **Qwen** on 140k Guitar Pro files; achieved **84.47% TAB accuracy** and **99.96% pitch consistency** on 14,368 tracks test set containing 15.1m notes.
- Engineered an end-to-end guitar accompaniment TAB generation pipeline that transforms structured musical data into performance-ready MusicXML scores; generated **528 arrangements for 176 songs across three difficulty levels**, with **148 passing tests** and zero export failures.
- Built and deployed a password-protected **internal full-stack platform** for score review and version management, supporting drag-and-drop file/folder uploads, deletion, and distribution of more than **1,400 generated score files**.

Data Science Intern, Leizhi Information Technology, Shanghai, China

July - August 2024

- Participated in **data analysis of Geographic High-Risk Merchants** for a government agency in Pudong, Shanghai; developed **GIS-based geographic heat maps** to support government regulatory oversight.
- Participated in the development of a **real-time complaint and reporting monitoring system** for a government agency, providing decision support for government management.

Audio Interface Designer, Huaxin Musical Instrument, Los Angeles, CA

January - May 2024

- Designed a **soundboard-style audio interface** in Blender for live streaming and recording with electronic piano and drum kits; specified three preamp channels, XLR and line inputs, VST-compatible audio drivers, multiple sample rates and bit depths, USB-C connectivity, and electrical parameters optimized for performance and affordability.

RESEARCH EXPERIENCE

Research Seminar, Optical Music Recognition for Jazz Lead Sheet: [Github](#)

August 2025 - present

- Proposed **end-to-end system for full-page handwritten jazz lead sheet recognition**, leveraging a Sheet Music Transformer to jointly transcribe melody and chord symbols from images.
- Designed a **9-stage curriculum learning strategy** to progressively scale from single-staff inputs to full-page transcription using ~1.8k samples.
- Developed a detect-and-concatenate baseline using YOLO-based staff detection followed by staff-level recognition and reconstruction, and introduced chord-focused evaluation metrics.
- Achieved **substantial improvements** on a 32-page handwritten test set, reducing chord symbol error rate from 47.8% to 30.7% ($p < 0.001$), significantly outperforming the baseline.

Independent Research, Real-Time Human-Robot Improvisation System: [Github](#)

January 2026 - Present

- Developed a **real-time human-robot improvisation system** based on the Anticipatory Music Transformer (AMT), extending it from offline MIDI continuation to interactive co-performance with a marimba-playing robot.
- Fine-tuned AMT on the PiJAMa jazz improvisation MIDI dataset to better align generated output with jazz improvisational style and human musical interaction.
- Designed a real-time pipeline where human MIDI input is streamed via OSC to a cloud GPU for inference and Viterbi decoding, then transmitted back for robotic actuation.

Undergraduate Thesis, Music Dept. Senior Comprehensive Project: [Github](#), [PDF](#)

June - December 2024

Tracing the Evolution of Rock and Popular Music Harmony: A Computational Study Using Shannon Entropy

- **Designed and executed** a computational musicology study analyzing 890 rock songs (1950s–2020s) from the McGill Billboard dataset.
- **Implemented Shannon entropy** as harmonic complexity metric, demonstrating the viability of information theory as a quantitative framework for genre evolution research.
- **Discovered a measurable rise and fall in harmonic complexity**, peaking in late-70s progressive rock before declining in the 1980s and diversifying through the 1990s–2000s
- **Presented** results at Occidental College's Music Department Senior Project Conference.

Undergraduate Thesis, Computer Science Dept. Senior Comprehensive Project: [Github](#), [PDF](#) June - December 2024

Reference-free metric for evaluating audio stem separation quality

- **Developed a novel reference-free evaluation method** for audio stem separation systems, eliminating reliance on original ground-truth stems.
- **Designed and implemented two new metrics** — Frequency Isolation Score (FIS) for spectral leakage detection and Dynamic Stability Score (DSS) for temporal consistency analysis.
- **Evaluated on the MUSDB18-HQ dataset** (150 professionally mixed songs) and demonstrated strong alignment over traditional SDR/SIR/SAR metrics.
- **Presented** results at Occidental College's Computer Science Senior Project Conference.

MUSIC RECORDING AND PRODUCTION

- [Senior Comprehensive Project Album](#): Player; Recording, Mixing, Mastering Engineer; 11 songs (2024)
- [Occidental College Chamber Jazz Album 2024](#): Player; Recording, Mixing, Mastering Engineer; 11 songs (2024)
- [Occidental College Chamber Jazz Album 2023](#): Player; Recording, Mixing, Mastering Engineer; 10 songs (2023)
- [Occidental Music YouTube Channel](#): edited the audio and visual component of 15 recorded performances and published them on the Music Department's official YouTube channel (2022)
- ["Berret Yuffee's EP"](#) (three songs composed by an Occidental student): recorded vocal and guitar; mixed and mastered in Pro Tools (2024)

MUSIC STUDIO INDUSTRY EXPERIENCE

Production Manager Assistant, Occidental College Music Department, Los Angeles, CA *January 2022 – May 2025*

- Assist with event promotion, logistics, and social media management for on-campus live musical performances (15-20 events per semester).
- Live Stream and Record events on [YouTube](#), including Set up/Tier down live stream equipment and monitor the soundboard, OBS streaming system, and other audio equipment.
- Manage inventory (cameras, microphones, speakers, soundboards, cables); create a labeling system for equipment, and maintain a digitized archive of past performance materials.

Studio Assistant, Occidental College Choi Studio Assistant, Los Angeles, CA *August 2023 – May 2025*

- Manage Weekly inventory (worth \$60K+), including checking in and out equipment, monitoring equipment status after returns; maintain studio and storage space, Install and ensure software is ready for use.
- Supervise and Provide technical support for students and enforce the studio's leasing policies (2-3 three hour session per week)
- Designed and managed EMS(Electronic Music Studio) at Occidental College including installing audio equipment, manage software update, and coordinate work orders cross departments (Music, IT, and Maintenance)

Recording Intern, Kingsize Soundlabs, Los Angeles, CA *February – June 2023*

- Shadowed professional engineers in recording sessions, including collaborations with Lyla Forde, Will Brahm, Walter Trout, Aidan Carroll, Julius Rodriguez, Drown, and Dudley.
- Assisted in studio operations, including assembling and disassembling recording equipment, daily studio cleanup and maintenance, managing inventories, maintaining the single VIP vocal studio, and coffee and lunch runs.

Music Production Intern, Jiali Music Studio, Shanghai, China *April – June 2021*

- Prepare client meetings to learn about clients' needs for OST, commercial, video game music

LEADERSHIP & SERVICE

- **Vice President**, Georgia Tech Audio Engineering Society (AES) Club — Organize mix-feedback sessions, train students to use Georgia Tech's recording studios and audio equipment, and coordinate guest lectures with industry professionals and researchers.
- **Treasurer**, Occidental Chinese Students & Scholars Association — Managed an annual budget of **\$5,000** and organized Lunar New Year celebrations, biweekly movie nights, concerts, bonfires, and new-student social events.
- **Student Representative**, Suno AI at Occidental College

SKILLS & INTERESTS

- Programming: Java, Python, C/C++, HTML/CSS, MAX/MSP, SQL, Blender, Latex, Mathematica
- Music production: DAWs (Logic Pro X, Ableton Live, Pro Tools), Izotope RX, Izotope ozone, Melodyne
- Instruments: Saxophone (10+years), Guitar, Piano, Tuba, and Drums
- Language: Chinese Mandarin (Native) and English (Fluent)
- Interests: e-sport (League of Legends North America top 20%), Oxy Intramural Basketball Team, Snowboarding (black diamond, freestyle park)